LUCA TEODORI

Curriculum Vitae

Instituto de Astrofisica de Canarias, La Laguna (Spain)
✉ *luca.teodori@iac.es*
My Webpage
Github

Summary

I am a postdoc in theoretical astrophysics. My main interest involves the use of gravitational probes, most notably gravitational lensing and galactic stellar kinematics, and how we can use them to uncover the nature of dark matter and the structure of galaxies. I am also interested in cosmology (large scale structures, early universe) and on fundamental physics. I mainly worked in degeneracies in strong gravitational lensing, and on the Ultralight dark matter model. Various codes (mostly in C++ and python) I used/developed for my projects can be found on my github page, `https://github.com/lucateo`
Nationality: Italian. Email address: luca.teodori@iac.es

Education

2025- **Postdoc**, *Instituto de Astrofisica de Canarias*, Spain.

June 2025 **PhD degree in physics**, *Weizmann Institute*, Israel.
Thesis Title: "Gravitational Probes of Ultralight Dark Matter and Degeneracies in Time-Delay Cosmography". Supervisor: Prof. Kfir Blum

Publications

Journal Articles

2604.01278 **Andrea Caputo and Luca Teodori**, *"Influence of tides and self-gravity on Ultra-Light Dark Matter Bounds from Dwarf Galaxies"*, Published in: Phys.Rev.D 114 (2026) 6, 063013.

An updated list of my publications can be also found in https://inspirehep.net/authors/1865293

2602.02802 **Luca Teodori**, *"The approximate gravitational lensing multiple plane mass sheet degeneracy"*, Published in: JCAP 05 (2026) 046.

2504.16202 **Kfir Blum, Marco Gorghetto, Edward Hardy and Luca Teodori**, *"Bracketing the soliton-halo relation of ultralight dark matter"*, published in: JCAP 06 (2025) 050.

2501.07631 **Luca Teodori, Andrea Caputo and Kfir Blum**, *"Ultra-Light Dark Matter Simulations and Stellar Dynamics: Tension in Dwarf Galaxies for $m < 5 \times 10^{-21}$ eV"*, Published in: Phys.Rev.D 113 (2026) 2, 023055.

2409.04134 **Kfir Blum and Luca Teodori**, *"AxionH0graphy: hunting for ultralight dark matter with cosmographic H_0 bias"*, published in: Phys.Rev.D 111 (2025) 4, 043509.

2305.19151 **Luca Teodori and Kfir Blum**, *"Host group degeneracy in gravitational lensing time delay determination of H_0"*, published in: JCAP 08 (2023) 065.

2201.05111 **Luca Teodori, Kfir Blum, Emanuele Castorina, Marko Simonović and Yotam Soreq**, *"Comments on the mass sheet degeneracy in cosmography analyses"*, published in: JCAP 07 (2022) 07, 027.

2106.05976 **Simon Birrer, Anowar J. Shajib et al**, *"lenstronomy II: A gravitational lensing software ecosystem"*, published in: J.Open Source Softw. 6 (2021) 62, 3283, J.Open Source Softw. 6 (2021) 3283.

2105.10873 **Kfir Blum and Luca Teodori**, *"Gravitational lensing H0 tension from ultralight axion galactic cores"*, published in: Phys.Rev.D 104 (2021) 12, 123011.

Submitted

2605.18182 **Luca Teodori, Kfir Blum and Zhaouy Bai**, *"Gravitational lensing time delay beyond the Shapiro/geometry split"*, Submitted to A&A.

Workshop/Conferences participation

June 2026 **Montpellier Strong Gravitational Lensing Workshop 2026**, Montpellier.
Participation as speaker

May 2026 **SCALE: Strings & Cosmology – All Lengths Explored**, La Arena, Tenerife.
Participation as organizer

December 2025 **New opportunities to detect gravitational waves and dark matter in orbital dynamic**, Benasque.
Participation as speaker

November 2025 **Light Dark Sectors @ LNF**, Frascati.
Participation as speaker

December 2024 **Let there be light (particles) Workshop**, Daejeon.
Participation as speaker

May 2024 **Strong gravitational lensing beyond the main lens**, Montpellier.
Participation as speaker

April 2024 **IPS conference**, Tel Aviv.
Participation and short talk in the 2024 Israel Physics Society conference.

March 2024 **Dark Matter Beyond the Weak Scale II**, Durham.
Participation as speaker

November 2023 **New Physics from Galaxy Clustering II**, Trieste.
Participation and short talk in the IFPU workshop

April 2023 **IPS conference**, Tel Aviv.
Participation and poster presentation in the 2023 Israel Physics Society conference.

February 2022 **IPS conference**, Be'er Sheva.
Participation and short talk in the 2022 Israel Physics Society conference.

Visiting

October 2025 **Visit IFAE**, Barcelona.
Visit to IFAE, host: Prof. Diego Blas

July 2025 **Visiting CERN**, Geneva.
Visit to CERN, host: Prof. Andrea Caputo

December 2024 **Visiting DESY**, Hamburg.
Visit to DESY Hamburg, host: Dr. Marco Gorghetto

November 2024 **Visiting KIAS**, Seoul.
Visit to KIAS, host: Prof. Seung Joon Lee

Jan/Feb 2024 **Visiting Perimeter Institute**, Waterloo.
Visit to Perimeter Institute, host: Prof. Sergey Sibiryakov.

December 2023 **Visiting Milano University**, Milano.
Visit to Milano University, host: Prof. Emanuele Castorina.

October 2023 **Visiting ICTP**, Trieste.
Visit to ICTP, host: Prof. Giovanni Villadoro.

June 2022 **Visiting CERN**, Geneva.
Visit to CERN, host: Dr. Marko Simonović.

October 2021 **Visiting Pisa University** , Pisa.
Visit to Pisa University, host: Prof. Paolo Panci.

Aug-Dec 2020 **DESY**, Hamburg.
Actual start of the PhD in DESY Hamburg, thanks to my affiliation to the Multimessenger Astronomy School. Supervision of Prof. Kfir Blum. Host: Prof. Rafael Porto.

Participation to schools

January 2025 **GGI Lectures on the Theory of Fundamental Interactions 2025**, Florence.
Participation to the GGI school 2025.

June 2023 **Summer School on Particle Physics**, Trieste.
Participation in the ICTP Summer School on Particle Physics 2023.

November 2022 **XXXIII Canary Islands Winter School of Astrophysics**, La Laguna.
Participation and short talk in the XXXIII Canary Islands Winter School of Astrophysics: Overlaps at the Frontiers of Astrophysics, Cosmology and Particle Physics.

July 2022 **Cargese 2022 Summer School**, Cargese.
Participation and short talk in the Cargese 2022 International Summer School: Rethinking Beyond the Standard Model.

September 2019 **Heraeus Summer school**, *Friedrich-Schiller-Universität*, Jena.
Participation and talk in the 2019 Heraeus Summer school on gravitational lensing.

Prizes

February 2024 **Chaim Mida travel prize**, *Prize for outstanding students in theoretical physics*, Weizmann Institute.

Teaching Assistantship

2024 **Quantum Field Theory**, *Weizmann Institute*, master course.

2023/2024 **Cosmology 2**, *Weizmann Institute*, PhD course.

2022/2023 **Quantum mechanics 1**, *Weizmann Institute*, master course.

2016/2017 **Tutor in geometry and linear algebra**, *Alessandro Volta College, Pavia*, bachelor course.

Educational qualifications

October 2019 **Master Degree in physics**, *Università degli studi*, Padova, 110/110 cum laude.
Thesis Title: "Non-linear evolution of cosmological power-spectra using the Kinetic Field Theory approach"
Supervisor: Matarrese Sabino
Co-supervisor: Bartelmann Matthias

September 2017 **Bachelor degree in physics**, *Università degli studi*, Pavia, 110/110.
Thesis Title: "A common framework for entropy in thermodynamics and quantum information theory"
Supervisor: Perinotti Paolo

Computer skills

Systems **Unix/Linux, Windows**.
Good knowledge of the bash environment.

Languages **C++, Python, PyTorch, LaTeX, html**.
Good knowledge of the C++ and python languages. Various C++ and python projects can be found on my github page, `https://github.com/lucateo` whereas my site (hosted on github servers) is in `https://lucateo.github.io/`